UNIVERSITY OF BRISTOL

Practice Paper

School of Computer Science

First in-class test for the Degree of Master of Science in Computer Science (conversion)

COMSM1302
Overview of Computer Architecture

TIME ALLOWED:
2 Hours

This paper contains **fifteen** questions.
All questions will be marked.
The maximum for this paper is **100 marks**.

Other Instructions

1. **Make sure you carefully read any instructions on the question paper and answer booklet.**
2. **You must use the answer booklet to answer all questions.**
3. **No calculators may be used for this examination.**
4. **This is a closed-book examination; books, notes, or other resources are forbidden.**
5. **A reference sheet will be given out alongside the question paper.**
6. **Any of the question paper can be used for rough work, but only the answer booklet will be marked.**

TURN OVER ONLY WHEN TOLD TO BEGIN WORK

Section 1 – Short-answer questions (50 marks total)

You do not need to justify your answers for any of the questions in this section, and you will not receive partial credit for showing your reasoning. Just write your answers down in the shortest form possible, e.g. "A" for multiple-choice questions, "true" for true/false questions, or "23" for numerical questions. If you do display working, circle or otherwise indicate your final answer, as if it cannot be identified then the question will not be marked.

Question 1 (4 marks)

Mark each of the following statements true or false.

(a) $(A \wedge B) \vee (C)$ is in disjunctive normal form. **(1 mark)**

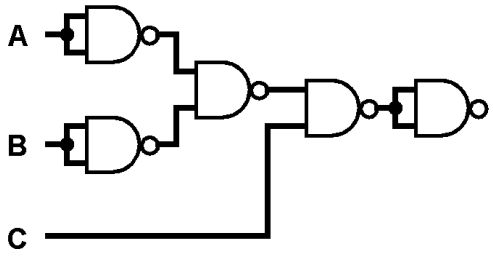
(b) $(B \vee \neg A) \wedge (C \vee A)$ is in disjunctive normal form. **(1 mark)**

(c) $\neg(A \vee B) \vee (\neg B \wedge C)$ is in disjunctive normal form. **(1 mark)**

(d) $(C \wedge \neg A \wedge B) \vee (B \wedge \neg B)$ is in disjunctive normal form. **(1 mark)**

Question 2 (4 marks)

Consider the below NAND circuit diagram.



The following Boolean expression is logically equivalent to the NAND circuit.

(A <OP1> B) <OP2> C

(a) Which logical operator should replace <OP1>? **(2 marks)**

(b) Which logical operator should replace <OP2>? **(2 marks)**

Question 3 (5 marks)

Consider the Hack ALU specification.

(a) What function is carried out, if the control bit `f` is set to 0? **(1 mark)**

(b) What happens to input `x`, if the control bits `zx` and `nx` are both set to 1? **(2 marks)**

(c) What will `out` compute, if the controls are set as `zx` $= 0$, `nx` $= 1$, `zy` $= 1$, `ny` $= 1$, `f` $= 0$, and `no` $= 1$? **(2 marks)**

Question 4 (5 marks)

Mark each of the following statements true or false.

(a) A latch is level-triggered. **(1 mark)**

(b) It is faster to access data stored in RAM than in ROM. **(1 mark)**

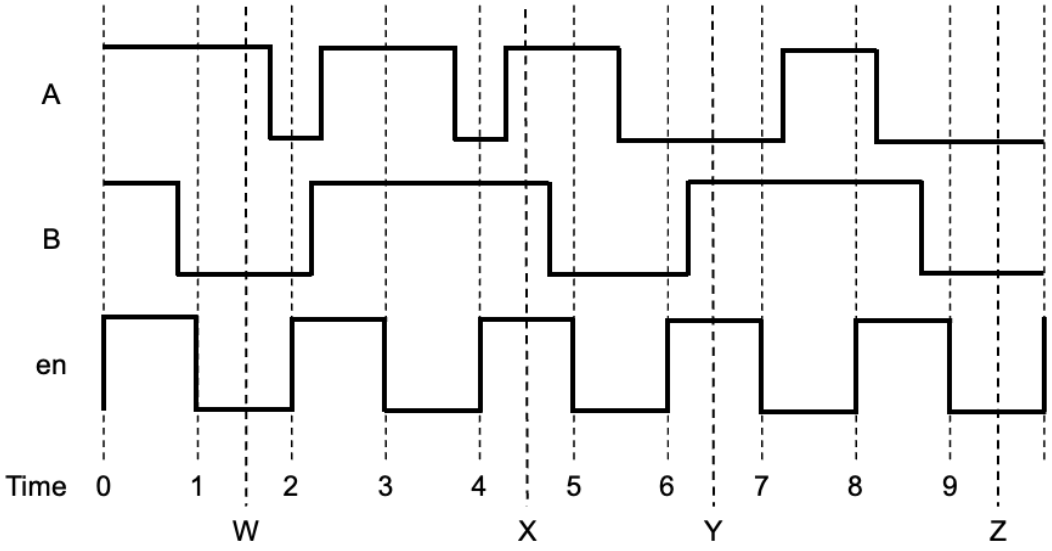
(c) A Mealy machine requires more states than the equivalent Moore machine. **(1 mark)**

(d) A demultiplexer with 2 select bits will select between 4 inputs. **(1 mark)**

(e) Given a low input, a one-shot outputs high for exactly one clock cycle. **(1 mark)**

Question 5 (4 marks)

Consider the below negative edge-triggered D flip-flop waveform, and the corresponding outputs `Q` and `Q'`.



The flip-flop's `D` input is connected to the signal `A XOR B`, which updates immediately after the value of `A` or the value of `B` changes.

(a) At time point W, what is the value of the flip-flop's output `Q`? **(1 mark)**

(b) At time point X, what is the value of the flip-flop's output `Q`? **(1 mark)**

(c) At time point Y, what is the value of the flip-flop's output `Q'`? **(1 mark)**

(d) At time point Z, what is the value of the flip-flop's output `Q'`? **(1 mark)**

Question 6 (4 marks)

Which of the following statements comparing memory is **false**?

A. An SRAM chip has lower capacity than a DRAM chip of the same physical size.

B. A flip-flop provides faster access speeds than an SRAM cell.

C. An SDRAM cell is a type of DRAM cell, and has nothing to do with SRAM.

D. A DRAM cell is more expensive to build than a flip-flop.

Question 7 (6 marks)

Consider the 8-bit binary number 1001 0010.

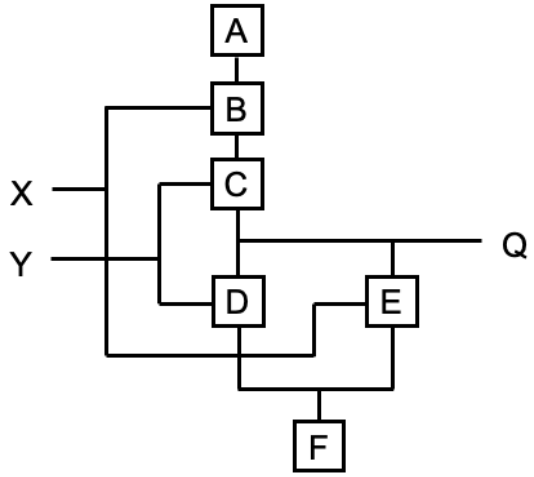
(a) What is its decimal value, if it uses unsigned representation? **(2 marks)**

(b) What is its decimal value, if it uses signed-magnitude representation? **(2 marks)**

(c) What is its decimal value, if it uses 2's complement representation? **(2 marks)**

Question 8 (6 marks)

Consider the below diagram of a NAND gate composed of transistors.



(a) Which voltage is connected at the marked location A? **(1 mark)**

(b) Which type of MOS transistor is connected at the marked location B? **(1 mark)**

(c) Which type of MOS transistor is connected at the marked location C? **(1 mark)**

(d) Which type of MOS transistor is connected at the marked location D? **(1 mark)**

(e) Which type of MOS transistor is connected at the marked location E? **(1 mark)**

(f) Which voltage is connected at the marked location F? **(1 mark)**

Question 9 (6 marks)

For each of the following clock time periods, give the equivalent clock frequency, in hertz.

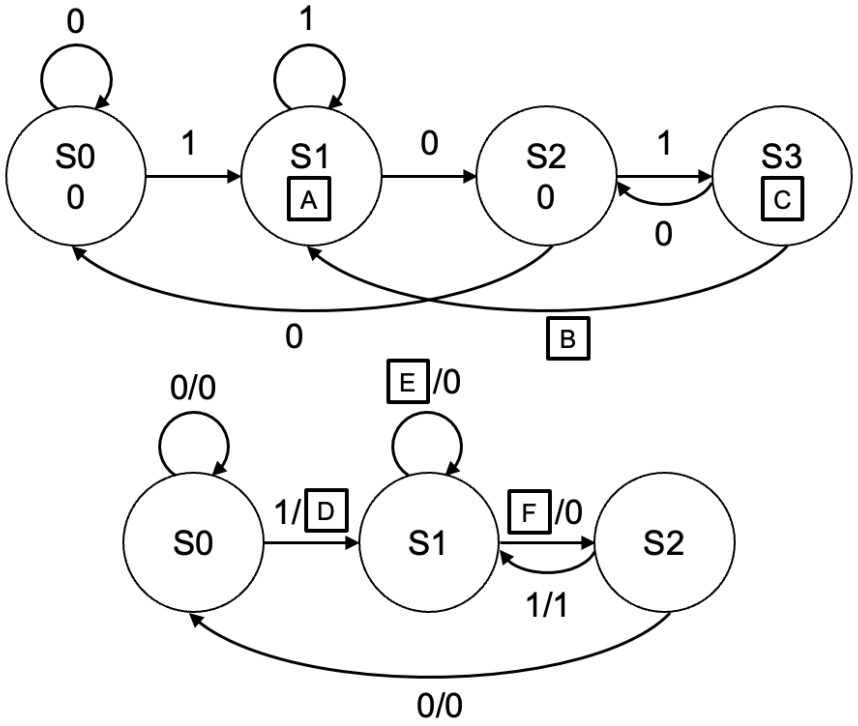
(a) $25ms$ **(2 marks)**

(b) $500\mu s$ **(2 marks)**

(c) $4s$ **(2 marks)**

Question 10 (6 marks)

Consider the below state transition diagrams, each representing the same finite state machine.



(a) Will the value at the marked location `A` be 0, 1, or impossible to determine? **(1 mark)**

(b) Will the value at the marked location `B` be 0, 1, or impossible to determine? **(1 mark)**

(c) Will the value at the marked location `C` be 0, 1, or impossible to determine? **(1 mark)**

(d) Will the value at the marked location `D` be 0, 1, or impossible to determine? **(1 mark)**

(e) Will the value at the marked location `E` be 0, 1, or impossible to determine? **(1 mark)**

(f) Will the value at the marked location `F` be 0, 1, or impossible to determine? **(1 mark)**

Section 2 – Long-answer questions (50 marks total)

In this section, you should give the reasoning behind your answers unless otherwise specified. You will receive credit for partial answers or for incorrect answers with sensible reasoning.

Question 11 (10 marks)

Consider the below truth table.

A	B	C	Out
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

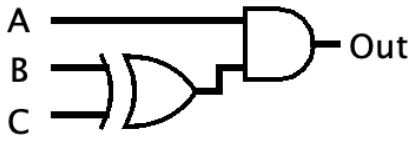
(a) Write a Disjunctive Normal Form (DNF) Boolean expression for the truth table, where each term represents exactly one row of the truth table. **(3 marks)**

(b) i. Draw a Karnaugh map that represents the truth table with boxes around appropriate groups of 1s, such that it will give a DNF Boolean expression with as few terms and variables per term as possible. **(4 marks)**

ii. Write a DNF Boolean expression that is given by your answer to part (b)(i). **(3 marks)**

Question 12 (13 marks)

Consider the below logical circuit.



(a) Write a Boolean expression that directly corresponds to the circuit, using a combination of $\wedge$, $\vee$, and $\neg$ operators only. **(3 marks)**

(b) Draw a truth table for the circuit, which shows all possible input combinations and their corresponding output values. **(4 marks)**

(c) Draw a logically equivalent circuit, which only uses 2-input NAND gates. **(6 marks)**

Question 13 (7 marks)

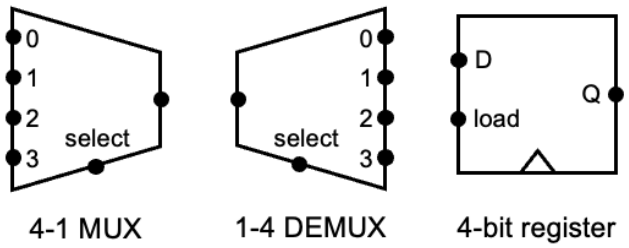
Consider a **4-bit adder** constructed from four 1-bit full adders, with 4-bit inputs `X` and `Y`, and a 1-bit input `C_in`.

(a) Write a Boolean expression for a **1-bit full adder**'s output `C_out`. **(3 marks)**

(b) Explain how this arrangement can be modified to calculate `X - Y`, referring to specific inputs of each full adder. **(4 marks)**

Question 14 (10 marks)

Draw a circuit for a **4-word RAM**, which takes a 4-bit input `in`, a 2-bit input `address`, a 1-bit input `load`, a 1-bit `clock` input, and produces a 4-bit output `out`. Only components shown below, as well as copies of them, are permitted.

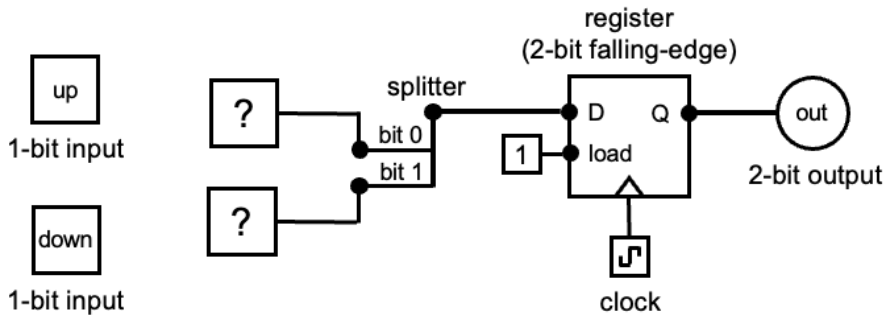


4-1 MUX 1-4 DEMUX 4-bit register

You may draw components in any orientation or style, as long as its inputs and outputs are unchanged. Label every component you use, along with its inputs and outputs.

Question 15 (10 marks)

A partial circuit is shown below, with 1-bit inputs `up` and `down`, and 2-bit output `out`. The boxes containing `?` each represent a missing part of the circuit. Wires drawn with a thick line carry a 2-bit signal, while wires drawn with a thin line carry a 1-bit signal.



The inputs `up` and `down` will never be 1 at the same time, and on a falling clock edge:

- If the input `up` is 1, the value of `out` should increase by 1 e.g. if `out` was previously 00 then it should now be 01. However, if `out` was previously 11 then it should stay 11.
- If the input `down` is 1, the value of `out` should decrease by 1 e.g. if `out` was previously 11 then it should now be 10. However, if `out` was previously 00 then it should stay 00.

The circuit's signals are labelled as follows:

Variable	Meaning
up	the input `up`
down	the input `down`
A	MSB of the register's output `Q`
B	LSB of the register's output `Q`

(a) Give a Boolean expression for the `bit0` input of the splitter. **(5 marks)**

(b) Give a Boolean expression for the `bit1` input of the splitter. **(5 marks)**

END OF EXAMINATION

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